

MULTI-TENANT BUSINESS OPERATING SYSTEM

# BusinessOS

## Full Project Presentation

Spring Boot 4 · Angular 22 · PostgreSQL · Multi-Provider AI

Prepared from the current codebase — architecture, modules, and status as implemented today.

# One Platform, Every Business Operation

- BusinessOS is a single, multi-tenant SaaS that replaces a stack of disconnected point tools — each company ("tenant") gets isolated data across every module.
- Eleven backend modules cover the full operating surface: CRM, HRM, Finance, Service Desk, Support, ITAM, AI, Company/platform admin, Website (public CMS), Search, and Dashboard analytics.
- The Angular frontend mirrors this with ~19 feature modules, so each business function (sales, people, money, tickets, assets) gets its own dedicated workspace inside one login.
- Every entity that belongs to a tenant is isolated at the ORM layer via a Hibernate tenant filter keyed on `company_id` — verified consistently across CRM, HRM, Finance, and AI, not just bolted on to one module.

# Tech Stack

## Backend

- Java 21
- Spring Boot 4.1.0
- Hibernate / JPA
- PostgreSQL

## Frontend

- Angular 22 (standalone)
- Template-driven forms
- Bootstrap 5
- New @if/@for control flow

## Auth & Security

- JWT — 15-min access / 7-day refresh
- Email verification & reset flows
- Admin impersonation (audited)
- Subscription-plan access gating

## AI Gateway

- Default: Google Gemini 2.5 Flash
- Fallback: Claude, OpenAI, Groq
- Per-tenant usage quotas
- Full request/response audit log

Also integrated: bKash / Nagad / Rocket / SSLCommerz payment rails, SMTP email delivery.

# Multi-Tenancy & Security Model

## Tenant isolation at the ORM layer

A Hibernate `@Filter` (`tenantFilter`) scopes every query to `company_id` — applied to Lead, Opportunity, Client, Employee, Budget, Chart of Accounts, AI conversations, and more. Not a per-query afterthought.

## Permission checks on every mutation

Each service method calls `authorizationService.checkPermission(...)` with a granular `PermissionCode` (e.g. `EMPLOYEE_CREATE`, `OPPORTUNITY_UPDATE`) before acting — not just role-gated at the controller.

## JWT authentication with refresh rotation

15-minute access tokens, 7-day refresh tokens, email verification, and password reset flows.

## Admin impersonation, audited

Support/admin staff can act as a tenant user for troubleshooting — every impersonation session is logged.

## Subscription-plan enforcement

A `SubscriptionEnforcementFilter` gates feature/module access by the company's active plan.

# CRM — Sales & Customer Management

- Full pipeline: Lead -> Client/Account -> Opportunity, with a separate multi-contact-per-account model — the standard relational shape, not a flattened toy schema.
- Opportunity stages carry real default probabilities (Prospecting 10% -> Closed Won 100%); closing a deal credits the client's lifetime value, and reopening a won deal correctly debits it back out.
- Losing a deal requires a reason, fires an owner notification, and logs a system activity — stage changes have consequences, not just a status label flip.
- Pipeline forecasting is computed live (per-stage count, total value, probability-weighted forecast) — not hardcoded dashboard numbers.

# HRM — People & Payroll

- Employee lifecycle: hire -> probation -> confirmed -> active -> resigned/terminated, with a real org chart (self-referencing reporting manager, department head/parent hierarchy).
- Leave requests reserve a pending balance on submit and convert it to used days on approval — with correct balance reversal on cancellation depending on whether leave already started.
- Payroll computes attendance-based deductions from real absence records and posts a balanced double-entry transaction to the General Ledger on payment (Dr Salaries Expense / Cr Cash / Cr Payroll Payable).
- Recruitment now closes the loop end-to-end: an OFFERED candidate can be hired directly into a full Employee + portal-user record from the Applications screen — the one gap found in review has been closed this cycle.

# Finance — Accounting Suite

- Chart of accounts, general ledger, journal entries, budgets, and fixed assets — the full double-entry accounting core.
- Accounts payable side: vendor management, vendor bills, expenses, and accounting-period close controls.
- Bank reconciliation against statement lines, plus client invoicing with invoice items, credit notes, and refunds.
- A dedicated payments flow and a reports sub-module round out what is, after CRM and HRM, the most fully built section of the platform.

# Service Desk — Internal Workflow Engine

- Dynamic, form-driven service requests rather than a fixed ticket schema — request fields are configurable per service type.
- Multi-stage, configurable approval workflows with their own stage and template sub-packages, plus task tracking through each stage.
- Quotation submit/reject flow for priced service requests, SLA-style billing, a knowledge base, and post-resolution reviews.



# Support — Customer Helpdesk

- A distinct ticketing system from Service Desk — purpose-built for customer-facing support rather than internal workflow requests.
- Tickets carry priority, status, type, and source; agents have their own status tracking (available/away/busy).
- SLA policies, threaded ticket messaging, category tagging, and audit logging on every ticket action.

# ITAM — IT Asset Management

- Extends the core Asset entity (shared with HRM) with software license management — license types, seat allocation, and renewal tracking.
- Employee offboarding checklists specifically for asset return, so a termination isn't complete until hardware/licenses are reclaimed.
- Bulk asset import via CSV with row-level error reporting, plus full asset assignment history.

# AI Gateway

- A single internal gateway with pluggable provider adapters — Gemini (default, 2.5 Flash), Claude, OpenAI, and Groq are all wired in, so a provider outage or cost change doesn't require a rewrite.
- Per-tenant usage governance: daily company quota and hourly per-user quota, enforced before a request is sent.
- Every AI call is logged for audit — used today for lead/candidate summaries, employment-letter drafting, and a global "ask" search layer.

# Platform & Supporting Modules

## Company

The tenant root record — company registration/update, team invites, and per-company leave-policy configuration.

## Website

A CMS for the public marketing site — FAQs, nav, pricing plans, portfolio, and team profiles. Separate from the tenant application itself.

## Search

Cross-entity global search, plus an AI-assisted "ask" Q&A layer built on the AI gateway.

## Dashboard

KPI aggregation and platform-wide metrics (plan mix, growth over time) for both company dashboards and platform admin analytics — no entities of its own, pure aggregation.

# Recent Enhancements

## **Recruitment -> Employee conversion (closed a known gap)**

Added a full "Hire" action: an OFFERED job application can now be converted directly into an Employee record with a portal login, in one transaction — reusing the same onboarding path (welcome email, notification defaults) as a manual hire. Previously HR had to re-enter the employee by hand.

## **Brand color pass**

Rolled the platform's color tokens onto a Deep Navy / Electric Cyan / Light Slate palette with Emerald / Amber / Red status colors — applied through the shared theme tokens and the navy page-header component used across 30+ admin screens, without touching any existing card styling or layout.

# Known Limitations & Roadmap

## Post-close automation is a stub

Winning a CRM deal fires an event, but auto-creating a client-onboarding workflow from it is an explicitly-flagged extension point, not yet implemented.

## Biometric attendance matching is a placeholder

The fingerprint-matching path is a labeled stand-in for a real SDK integration; manual, GPS, QR, and RFID check-in are unaffected.

## Lead/duplicate detection is minimal

Only exact-email matching today — no fuzzy name/company matching yet.

## No field-level audit history

Activity timelines exist, but there's no "amount changed from X to Y" style change log beyond that.

### Why this matters:

None of these block day-to-day use — each is a scoped, clearly-bounded gap rather than a hidden shortcut, and each has an obvious next step.

## SUMMARY

# A real operating system, not a demo.

Eleven backend modules, tenant-isolated by default, with permission checks, notification flows, and business rules that hold up under scrutiny — CRM stage math, payroll GL postings, and now a closed-loop recruitment pipeline all reflect how the underlying business actually runs.

Thank you.